Does LIHTC Development Affect Local Housing Costs? A Bayesian Hierarchical Analysis

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Abstract

Housing policy has becomes a popular political topic in the U.S., and often revolves around market-rate housing. When affordable housing is addressed it's usually spoken about in a negative light with little empirical evidence. I wanted to understand the relationship between affordable housing, specifically LIHTC, and housing costs (both home and rent prices). I used a Bayesian hierarchical model with county intercepts and state-level slopes, and after controlling for population growth and deflating prices using CPI-less-shelter, I find a modest positive association of 0.13% for home values and 0.32% for rents per 1% increase in LIHTC per capita. I used a LOO comparison to compare a few different models and the framework decisively favors the inflation-adjusted hierarchical model. The positive association likely reflects LIHTC allocation patterns rather than a causal price effect, though our analysis indicates that housing policy is too variable across states to have a robust national solution.

1 Introduction

"Affordability" has become a buzzword in the political discourse, most notably housing affordability has become a massive political talking point as Americans are overwhelmingly less able to afford a home these days (NAHB, 2024) . Many political pundits only discuss public policy surrounding market rate housing, most notably their desire to lighten local zoning laws to allow for more market rate development. While I agree with this policy, I notice those pundits often have an aversion to affordable housing programs for a number of reasons. My goal is to understand how affordable housing programs (specifically LIHTC given its scale across the country) affect housing costs, both in the rental market and the housing market.

The Low-Income Housing Tax Credit (LIHTC) is the largest affordable housing creation program in the United States, responsible for over 3.7 million affordable housing units since its creation with the Tax Reform Act of 1986. The federal government (though local housing authorities) allocate $\approx$ \$10.5 billion a year to state and local governments for "the acquisition, rehabilitation, or new construction of rental housing targeted to lower-income households." (HUD, 2025).

I felt this project was great for a Bayesian approach because I wanted to use a hierarchical model to capture the variation across counties in a state while still recognizing counties within the same state should be considered different from counties in another state. As I describe later, a fully pooled model would treat counties are truly independent samples when in reality counties within a state would share some similarities, leading to their housing costs being partially correlated. I also wanted a posterior distribution to help us compare different models we ran upon the data.

2 Data

I collected data from a few different sources to get the number of LIHTC units built in a county, the typical home price and rent price of a county, infation indices, and county populations.

2.1 LIHTC Database

The Department of Housing and Urban Development (HUD) has a LIHTC database describing over 54k projects from 1987–2023. The projects are geocoded with FIPS codes which describes a state-county-census tract. This is what I used to connect counties across datasets. I then aggregated so each row is a number of low income units built per county per year. I had to exclude ≈20.9% of projects due to invalid county FIPS codes.

2.2 Zillow Home Value Index (ZHVI)

Zillow Research creates a Zillow Home Value Index (ZHVI) which represents the "typical" home value (dollar-denominated) for homes in the 35th–65th percentile range (Zillow, 2025a) for a given geographic level from 2000–present. I download the smoothed and seasonally adjusted ZHVI by county, and annualized the index by taking the mean of monthly values per county-year.

2.3 Zillow Observed Rent Index (ZORI)

Zillow Research also creates a rent index (ZORI) which is a dollar-denominated "typical" market-rate rent for a given geographic level (Zillow, 2025b). I used a similar methodology as above where I took an annual average of smoothed and seasonally adjusted ZORI. ZORI had much spottier coverage (755 counties with 5+ valid years).

2.4 County Population Estimates

I wanted to account for housing demand as this would affect increases in both home values and rent prices, so I pulled in census population data from the U.S. Census Bureau Population Estimates Program: 2000–2009 intercensal estimates (Census Bureau, 2010), 2010–2019 vintage estimates (Census Bureau, 2020), and 2020–2023 estimates via USDA ERS (USDA ERS, 2025).

2.5 Inflation Less Housing Costs

A major conflating factor in increasing housing costs is inflation, which would always cause prices to increase regardless of LIHTC units constructed, so I pulled CPI Values for "All Items Less Housing" from the Federal Reserve of St. Louis (Federal Reserve of St. Louis, 2026), and "annualized" the data by taking the average CPI value for a given year. I didn't want to include housing prices because this is the very relationship I'm trying to measure, so I needed to control for the increase in costs of everything else (i.e. how can I account for the U.S. dollar losing value).

2.6 Panel Construction

Across these 4 sources we ended up with two datasets:

- Housing Dataset: ∼14,000 county-years across ∼2,400 counties and 50 states (2000–2023) to compare LIHTC's relationship with home values.
- Rental Dataset: ∼2,500 county-years across ∼700 counties (2015–2023) to compare LIHTC's relationship with rental prices.

In both datasets, I decided to use the cumulative sum of LIHTC units per capita (1k residents) rather than the total units built in a given year. I used a cumulative sum because once these units are built, they're a part of the housing supply for future years, so any effect these additional units may have will persist further in time than the year the units were built. Secondly, I scaled the sum relative to the county's population because adding 100 units of housing to a county with 100k in population may look different than 100 units of housing to a county with 1M in population, so we should treat these on a relative scale.

3 Model Specification

Both of these datasets went through the same set of models through out this project, and below are the three models I will compare and discuss in this report:

3.1 Pooled Baseline

We have a pooled baseline which contains a single intercept, single slope, and no hierarchy. The formula is as follows:

$$\log(\mathrm{ZHVI}_{c,t}) = \beta_0 + \beta_1 \cdot \log(1 + \mathrm{LIHTC}^*_{c,t}) + \gamma \cdot \mathrm{PopGrowth}_{c,t} + \varepsilon_{c,t} \tag{1}$$

$$\varepsilon_{c,t} \sim \mathrm{Normal}(0, \tau^{-1}) \tag{2}$$

where $\mathrm{LIHTC}^*_{c,t}$ denotes cumulative LIHTC units per 1,000 residents in county c at year t.

For our priors I used: $\beta_0 \sim \mathrm{Normal}(12,\ \tau = 0.25)$, $\beta_1 \sim \mathrm{Normal}(0,\ \tau = 0.001)$, $\tau \sim \mathrm{Gamma}(0.001, 0.001)$, and $\gamma \sim \mathrm{Normal}(0, 0.001)$. I decided on a more informative prior for β_0 since we had a lot of data on the distribution of log(ZHVI), which centered around 12. The ZORI model followed the same structure but with a prior of $\beta_0 \sim \mathrm{Normal}(7,\ \tau = 0.25)$,

3.2 Hierarchical Model with Population Control

I have a hierarchial model with county intercepts and state-varying LIHTC slopes. The formula is as follows:

$$\log(\mathrm{ZHVI}_{c,t}) = \alpha_c + \beta_{s[c]} \cdot \log(1 + \mathrm{LIHTC}^*_{c,t}) + \gamma \cdot \mathrm{PopGrowth}_{c,t} + \varepsilon_{c,t} \tag{3}$$

$$\alpha_c \sim \mathrm{Normal}(\mu_\alpha,\ \tau_\alpha^{-1}) \tag{4}$$

$$\mu_\alpha \sim \mathrm{Normal}(12,\ \tau = 0.25), \quad \tau_\alpha \sim \mathrm{Gamma}(0.001, 0.001) \tag{5}$$

$$\beta_s \sim \text{Normal}(\mu_\beta,\, \tau_\beta^{-1}) \tag{6}$$

$$\mu_\beta \sim \text{Normal}(0,\, \tau = 0.001), \quad \tau_\beta \sim \text{Gamma}(0.001, 0.001) \tag{7}$$

Note: the ZORI model uses the same structure but with $\mu_\alpha \sim \text{Normal}(7, \tau = 0.25)$ and $\tau_\alpha \sim \text{Gamma}(2.0, 0.5)$ to address convergence with sparser data.

3.3 Hierarchical Model with Population and Inflation Control

While adding a hierarchical model improved the predictive ability of the model, inflation was a confounding factor as prices would always increase regardless of LIHTC units built, so I decided to take inflation into account. The formula and hierarchical model follow the above, except with index values being calculated as:

$$\text{Index}_{c,t}^{Real} = \text{Index}_{c,t}/(CPI_t/CPI_B)$$

Where CPI_B is the CPI of the earliest available year in the housing and rental datasets (2001 and 2015 respectively).

4 Results

4.1 Home Values (ZHVI)

Below are the model results for the final ZHVI Hierarchial Model

Table 1: Posterior Summary — ZHVI Hierarchical Model (Inflation-Adjusted, Normalized, Population-Controlled)

Parameter	Mean	SD	2.5% HDI	97.5% HDI	ESS Bulk	ESS Tail	$\hat{R}$
μ_α	11.497	0.013	11.472	11.522	612	1591	1.00
μ_β	0.130	0.020	0.090	0.168	4094	4121	1.00
τ_α	7.617	0.281	7.081	8.166	3818	5788	1.00
τ_β	56.136	12.016	33.692	79.623	6849	5556	1.00

Below are the traces for our parameters, forest plot for our state β and posterior distribution

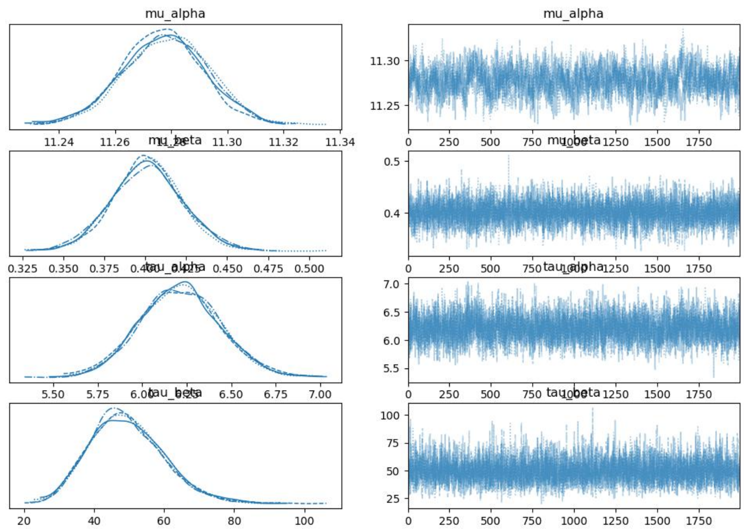

Figure 1: Traces for Bayesian Regression Parameters (ZHVI Model)

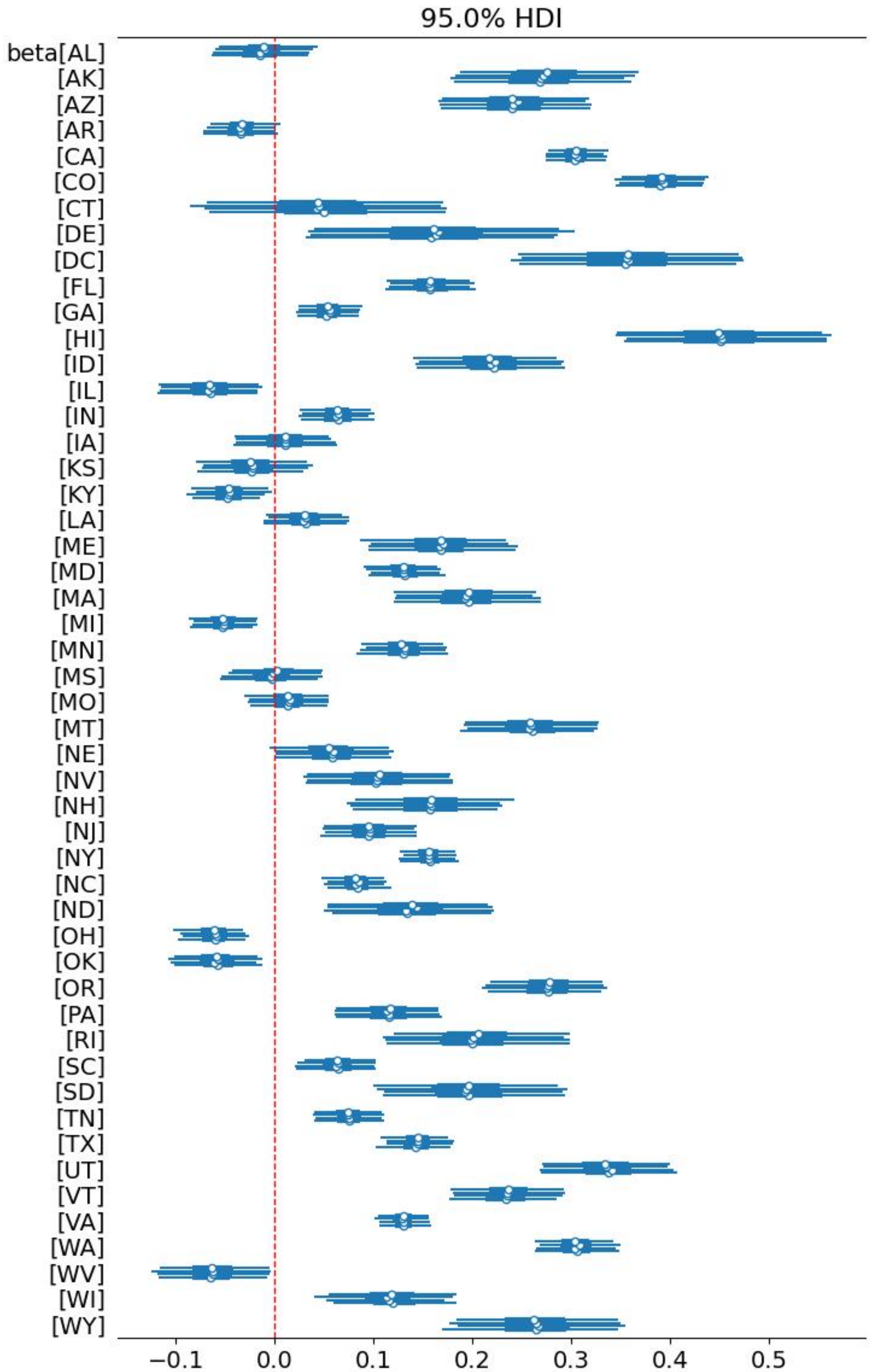

Figure 2: State-level LIHTC slope posteriors (ZHVI model). 95% HDI shown.

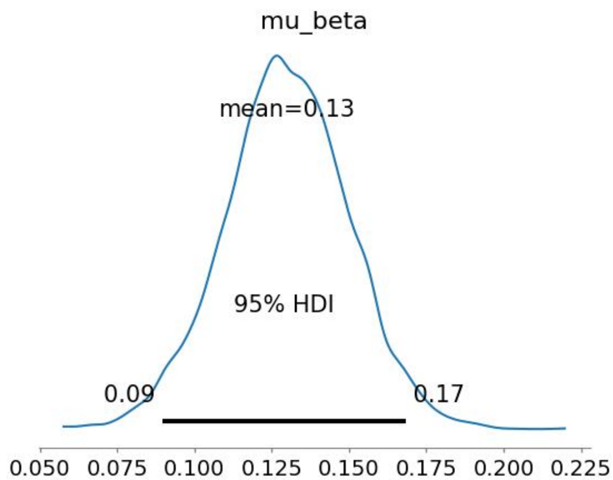

Figure 3: Posterior Distribution (ZHVI model). 95% HDI shown.

4.2 Rents (ZORI)

Below are the model results for the final ZORI Hierarchial Model

Table 2: Posterior Summary — ZORI Hierarchical Model (Inflation-Adjusted, Normalized, Population-Controlled)

Parameter	Mean	SD	2.5% HDI	97.5% HDI	ESS Bulk	ESS Tail	$\hat{R}$
μ_α	6.337	0.030	6.279	6.398	859	1549	1.01
μ_β	0.324	0.020	0.286	0.364	1510	3640	1.00
τ_α	11.295	0.779	9.830	12.884	2913	6862	1.00
τ_β	119.085	30.908	62.357	179.374	7610	10007	1.00

Below are the traces for our parameters, forest plot for our state β and posterior distribution

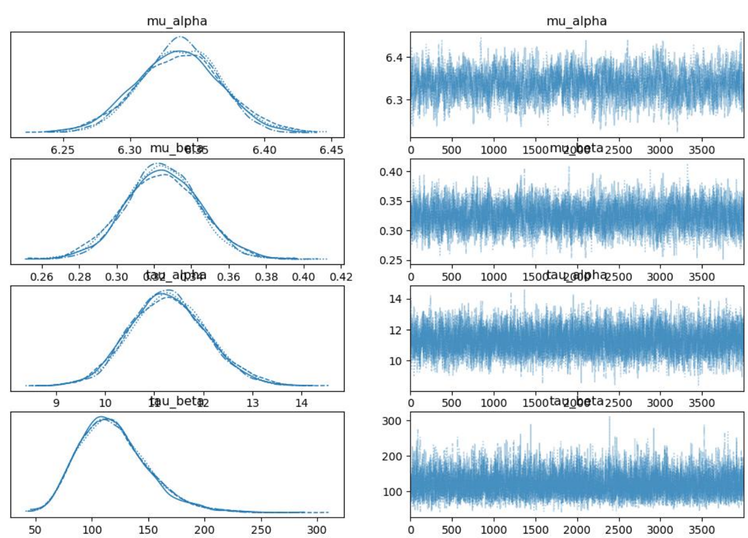

Figure 4: Traces for Bayesian Regression Parameters (ZORI Model)

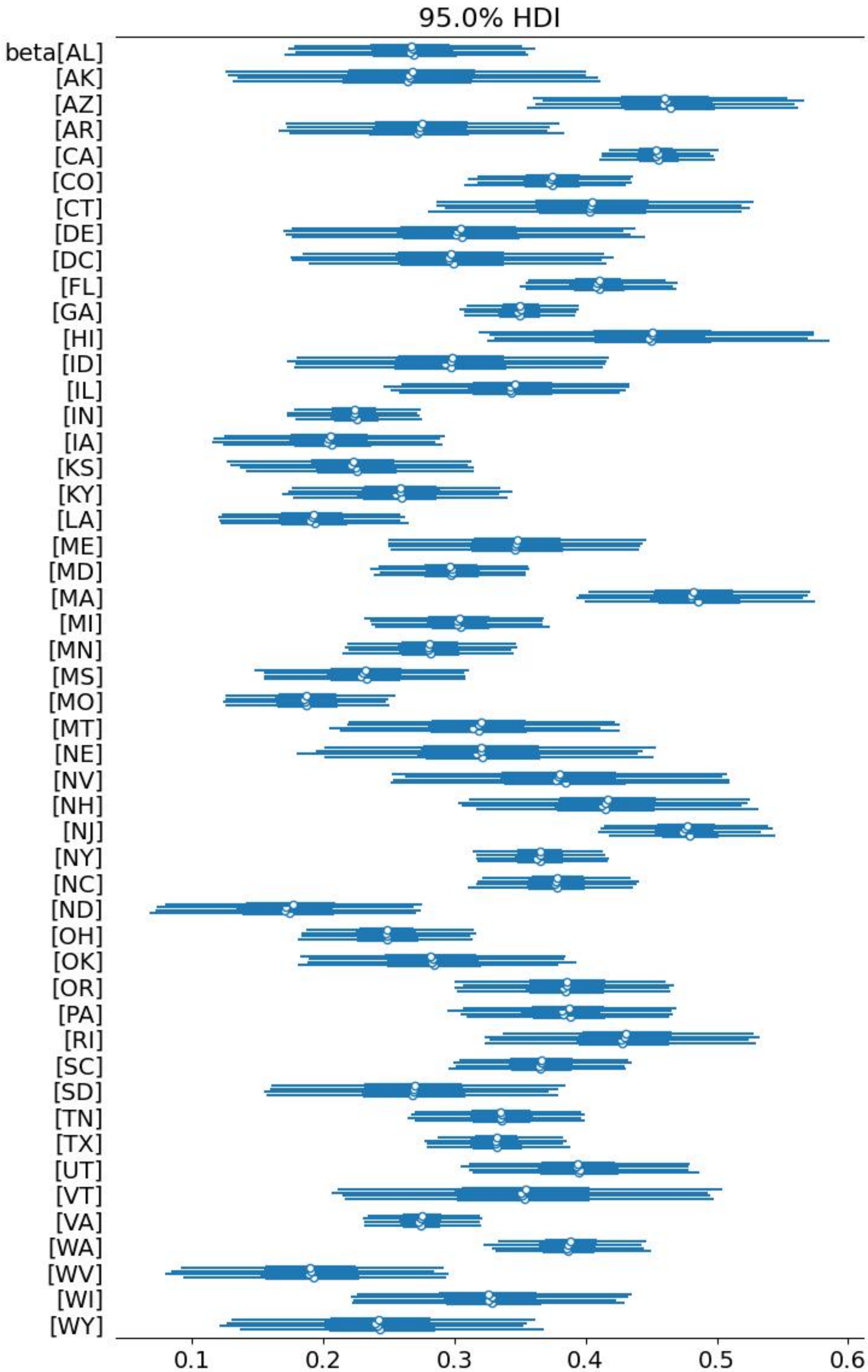

Figure 5: State-level LIHTC slope posteriors (ZORI model). 95% HDI shown.

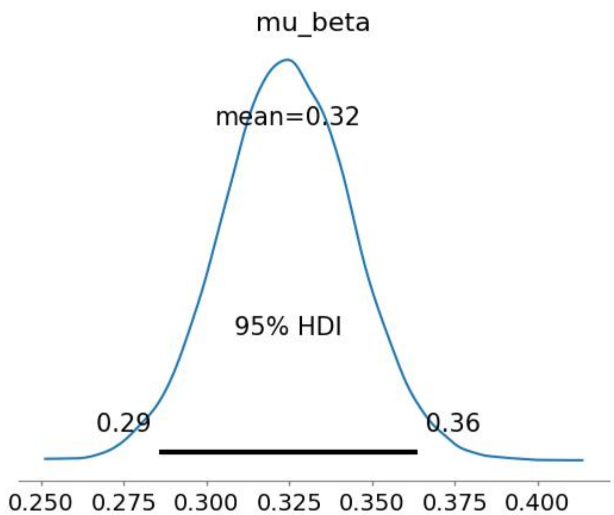

Figure 6: Posterior Distribution (ZORI model). 95% HDI shown.

4.3 Model Comparison

For both the ZHVI and the ZORI Models I compared their hierarchical model to a pooled version to see the lift in predictive power (i.e. does adding hierarchy help ascertain the relationship between LIHTC units per capita and housing costs). Below are the LOO Model comparisons for both ZHVI and ZORI:

Table 3: LOO Model Comparison — ZHVI (Home Values)

Model	Rank	ELPD LOO	p_{LOO}	Δ ELPD	Weight
Hierarchical (Inflation-Adj.)	0	3287.18	1632.72	0.00	0.996
Hierarchical (Nominal)	1	1688.17	1693.30	1599.01	$\approx$0.000
Hierarchical (No Norm.)	2	1669.94	1698.86	1617.24	0.004

Table 4: LOO Model Comparison — ZORI (Rents)

Model	Rank	ELPD LOO	p_{LOO}	Δ ELPD	Weight
Hierarchical (Inflation-Adj.)	0	2942.66	566.84	0.00	0.997
Hierarchical (Nominal)	1	1600.59	572.61	1342.07	$\approx$0.000
Pooled (Nominal)	2	-799.69	4.53	3742.35	0.003

5 Discussion

Across both the pooled and hierarchical models, we see a positive association between cumulative LIHTC units per capita and housing costs. Specifically, a 1% increase in cumulative LIHTC units per capita is associated with a .13% increase in home prices and a .324% increase in rental prices. The effect being stronger for rents than home values makes sense as LIHTC units are rental units, though what is counter intuitive is that somehow adding rental supply is correlated to increasing prices. All else equal, we would expect increasing supply to lower the rent cost. I think this shows there's still various confounding effects, and moreso reflects where LIHTC credits are allocated. Housing agencies are probably allocating these credits to areas with demonstrated housing need, which could be more than simply population growth, but things like job growth, wage growth, and zoning changes. States such as CA and IL having higher β_1 values than states like Alabama and Louisiana (even after controlling for population growth and normalizing LIHTC by population) indicate that we're still seeing a relationship tied to population and economic growth.

While I did not present the result of every model I tested in this report, the ZHVI and ZORI model had much higher β values when working with nominal index values. The association between LIHTC units and nominal home prices was 0.40%, suggesting a substantial portion of the raw relationship was driven by inflation trends rather than a direct relationship to LIHTC. We also see some of the state's 95% credible ranges contain 0 for the inflation–controlled model when they didn't in the nominal–controlled. These tend to be lower-cost and lower population states like Ohio and Oklahoma whereas the relationship is highest for states like Hawaii, Colorado, and CA.

There were a few limitations on the analysis. One of the bigger data limitations was ZORI coverage was limited to 2015 onwards, and many counties only had a couple years of data to work with. I think this is where our Bayesian model shined because we didn't need a lot of data to construct our posterior. We used the pooled data to construct our prior which helped the model focus on a specific range of values. Furthermore, the hierarchical structure ensures we could still discern county and state level effects even with limited data. LIHTC data also had missing county FIPS codes for $\approx 21\%$ of projects, which may have biased the data towards newer urban projects, but again our Bayesian model was well equipped to handle sparse data.

6 Conclusion

In conclusion we found that cumulative LIHTC units per capita have a positive association with home prices and rent prices (.13% and .324% respectively) even when trying to account for housing demand (measured in population growth). This association, however, varies greatly by state and shows the benefit of using a hierarchical approach to not pool all counties in the U.S. as independent samples but rather independent samples within their state where LIHTC units and housing regulations are often shared. The association between housing costs and LIHTC units are higher for rent price, which is sensible since LIHTC units are rental units, and we see the association stronger in states like CA where both population growth and housing demand are high, indicating the association is still highly linked to housing demand. Accounting for inflation reduced our 95% credible ranges and brought down the strength of the association, indicating much of the nominal relationship was simply linked to inflation. While none of the findings can be considered "causal", this does suggest housing policy would be best addressed at the state and local level rather than the national level to account for inter-state differences in housing demand and LIHTC effects on housing costs.

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